

DEL RIO, TX, USA

WMO: 722610

Lat: 29.378N

Lon: 100.927W

Elev: 999

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 22010

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.5	34.5	9.2	9.1	50.1	13.5	11.3	51.2	25.4	59.4	21.6	56.9	5.2	330	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.8	102.4	72.3	100.5	72.5	98.8	72.6	78.0	90.2	77.2	89.2	76.4	88.2	11.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	137.3	81.4	74.2	132.4	81.1	73.3	128.5	80.9	42.1	89.8	41.2	89.1	40.5	88.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.8	18.7	16.9	DB	24.7	106.1	3.6	2.5	22.1	107.9	20.0	109.4	18.0	110.8	15.4	112.6
			WB	22.1	79.9	3.4	2.3	19.6	81.5	17.6	82.9	15.8	84.2	13.3	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.3	53.0	58.1	65.2	72.5	79.4	85.0	86.8	87.0	81.1	72.1	61.3	53.3
	DBStd	13.95	7.84	8.93	8.12	6.83	5.76	3.95	3.81	3.62	5.15	7.16	7.83	7.44
	HDD50	153	59	28	6	0	0	0	0	0	0	1	10	49
	HDD65	1269	376	222	100	18	1	0	0	1	24	158	369	
	CDD50	7926	152	254	477	674	911	1051	1140	1147	933	687	349	151
	CDD65	3565	4	28	106	242	447	601	675	682	483	244	47	6
	CDH74	42756	73	310	913	2338	4959	7844	9199	9326	5367	2070	299	58
	CDH80	22330	10	85	289	971	2439	4335	5338	5438	2642	724	51	8
Wind	WSAvg	8.6	7.2	8.0	9.2	10.0	10.1	10.4	9.7	8.8	7.7	8.0	7.3	6.9
Precipitation	PrecAvg	19.0	0.6	0.9	0.9	1.9	2.2	2.2	1.6	1.5	2.4	2.3	1.0	0.6
	PrecMax	33.2	1.9	3.8	3.5	7.5	6.9	5.7	13.2	6.1	9.9	11.3	3.3	3.1
	PrecMin	9.3	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.3	0.5	0.9	0.9	1.7	1.5	1.8	2.4	1.5	2.2	2.5	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.8	88.3	91.5	98.0	103.0	104.9	104.8	104.1	102.3	94.2	84.8	79.6
		MCWB	58.8	59.7	61.2	65.4	68.8	72.2	72.9	73.3	71.4	70.8	65.4	60.2
	2%	DB	75.6	82.1	87.2	93.1	98.2	100.8	102.1	101.6	97.9	90.1	80.8	74.9
		MCWB	57.2	58.8	62.6	67.1	71.2	72.8	72.8	73.0	72.3	70.6	64.6	59.1
	5%	DB	71.4	77.6	82.9	89.3	94.8	98.3	100.1	99.9	94.9	87.2	77.4	70.8
		MCWB	55.4	59.1	62.4	66.8	70.7	72.8	72.8	73.0	72.2	69.8	63.4	57.9
	10%	DB	67.5	72.9	79.1	85.7	91.6	96.0	98.0	98.1	92.0	84.1	74.1	67.0
		MCWB	54.8	57.6	62.2	65.5	71.1	72.8	72.8	73.1	72.0	68.8	62.4	56.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	68.1	70.5	74.4	77.9	79.7	78.4	78.3	78.1	76.8	72.3	67.3
		MCDB	72.1	75.0	78.9	85.4	89.2	92.0	92.1	91.3	88.5	84.9	77.9	72.8
	2%	WB	63.2	65.9	68.9	72.7	76.4	78.1	77.3	77.2	76.8	75.0	69.5	64.6
		MCDB	69.0	72.2	76.8	83.0	87.3	90.0	90.3	90.6	87.1	82.2	74.8	69.0
	5%	WB	60.2	63.9	67.4	71.4	75.2	77.1	76.5	76.4	75.8	73.7	67.5	61.9
		MCDB	66.7	70.6	75.1	81.3	86.0	88.8	89.1	89.4	85.7	81.3	72.8	67.1
	10%	WB	56.8	61.4	65.9	70.2	74.2	76.1	75.5	75.5	75.0	72.4	65.5	58.4
		MCDB	64.3	69.3	74.0	79.5	85.0	87.6	87.7	88.1	84.7	80.0	71.9	64.7
Mean Daily Temperature Range	5% DB	MDBR	22.9	23.1	22.6	22.9	20.9	20.6	20.5	20.8	19.6	20.1	20.6	22.2
		MCDBR	30.5	29.6	27.9	27.3	25.6	23.6	23.1	22.9	22.9	22.5	24.4	28.0
		MCWBR	15.8	14.3	12.1	11.6	9.1	7.0	7.0	6.5	7.3	9.0	12.2	14.6
	5% WB	MCDBR	22.2	20.7	20.0	20.4	19.5	19.5	20.2	20.3	18.0	16.6	17.8	20.1
		MCWBR	14.0	12.7	10.5	10.1	7.9	6.8	6.4	6.2	6.2	7.6	11.2	13.5
Clear-Sky Solar Irradiance	taub	0.299	0.304	0.325	0.372	0.393	0.402	0.412	0.393	0.385	0.344	0.318	0.301	
	taud	2.558	2.551	2.503	2.384	2.382	2.400	2.408	2.463	2.427	2.539	2.536	2.552	
	Ebn at Noon	301	308	306	292	285	280	277	282	282	290	291	294	
	Edh at Noon	27	30	33	39	39	38	37	35	35	30	27	26	
All-Sky Solar Radiation	RadAvg	1053	1299	1624	1892	1961	2192	2167	2067	1701	1392	1111	957	
	RadStd	119	174	141	174	182	160	183	118	144	186	136	120	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page